

Carolyn Zeng

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Skills

Technical Skills: Python, R, SQL, Java, JavaScript, JSON, C, C++, Linux, Github Copilot

Certifications: AWS Certified Cloud Practitioner, MOS: Microsoft Excel Associate, MOS: Microsoft Access Expert

Speaking Languages: English (Native), Mandarin Chinese (Fluent), German (Intermediate)

Projects

Capstone Project (Children's Museum of Indianapolis)

Fall 2025

- Served as Data Lead and Analyst on capstone project for the Children's Museum of Indianapolis, delivering a Gradient Boost model in R that forecasts exhibit popularity with an R-squared of 0.52 given noisy real-world data
- Integrated multiple data sources (attendance records, foot traffic sensors, exhibit metadata) to identify key drivers of visitor engagement and optimize gallery space utilization
- Delivered actionable insights through comprehensive EDA and visualization dashboards, revealing exhibit lifecycle patterns and audience preferences to inform strategic planning

NYC Taxi Traffic Analytics (AWS Cloud Project)

Spring 2026

- Provisioned EC2 instances and developed ETL pipeline to process 2–10GB of NYC TLC trip data (Parquet format) stored in S3 with server-side encryption
- Orchestrated batch processing with Python and configured auto-scaling to enable scalable analysis of traffic patterns and congestion zones
- Delivered actionable insights identifying inefficient routes and high-congestion areas for potential route optimization

AI-Powered Vulnerability Detection

- Applied LLM-assisted code review and ML-based static analysis to identify security vulnerabilities in a software repository using Ollama in Python
- Exported structured findings (file, line, severity, confidence, mitigation) following industry schema for actionable security recommendations
- Prioritized detected vulnerabilities using severity/confidence framework to balance technical depth with clear communication

Experience

Alignerr — AI Training Data Contributor

July 2025 - Present

- Created and annotated structured datasets for AI/ML model training, contributing to 10+ projects with 90% labeling accuracy
- Provided programming and coding expertise as a member of the technical solutions group, resolving data creation issues and refining annotation guidelines
- Performed audio annotation tasks for natural language processing applications following strict quality standards

Education

Columbia University - Master of Science in Applied Analytics

May 2026

GPA: 3.66

College of DuPage - Data Analytics Certificate

In Progress

GPA: 4.0

University of Illinois at Urbana Champaign - Bachelor of Science in Astronomy
Computer Science Minor, Physics Minor

May 2024

GPA: 3.65